

20-2 OSPF Troubleshooting - Answer Key

In this lab you will troubleshoot the OSPF routing protocol.

OSPF Troubleshooting

- 1) OSPF has been configured on the network but the routers are not learning remote routes. The plan is to configure summarisation later but you need to fix basic OSPF first.

Your task is to troubleshoot the configuration and ensure that all routers have routes to every loopback and 10.x.x.x network.

There is a configuration error on each router. You will need to find and fix all errors before the network will work properly.

Note that some commands may be disabled during the CCNA exam so you should know the different troubleshooting commands.

You must not use the 'show running-configuration' command (or any derivative of it such as 'show run | section ospf') in this exercise.

There is not a set order of actions to troubleshoot this scenario. Troubleshooting in a logical fashion will however make it easier and quicker. This is how I would do it.

Check to verify that there are no OSPF routes on any router.

```
R1#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C      10.0.0.0/24 is directly connected, FastEthernet0/0
L      10.0.0.1/32 is directly connected, FastEthernet0/0
C      10.0.1.0/24 is directly connected, FastEthernet0/1
L      10.0.1.1/32 is directly connected, FastEthernet0/1
```

```
C      10.0.2.0/24 is directly connected, FastEthernet1/0
L      10.0.2.1/32 is directly connected, FastEthernet1/0
C      10.0.3.0/24 is directly connected, FastEthernet1/1
L      10.0.3.1/32 is directly connected, FastEthernet1/1
      192.168.0.0/32 is subnetted, 1 subnets
C      192.168.0.1 is directly connected, Loopback0
```

None of the routers have any OSPF routes.

'show ip protocols' shows that OSPF has been enabled on all routers.

```
R1#sh ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.0.1
  Number of areas in this router is 1. 1 normal 0 stub 0
  nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.255.255.255 area 0
    192.168.0.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)
```

On R1, 'show ip ospf interface' shows that all interfaces are in Area 0 (this information can also be seen in the 'show ip protocols' output), but it is meant to be an internal router with all interfaces in Area 1.

```
R1#sh ip ospf interface

FastEthernet0/1 is up, line protocol is up
  Internet address is 10.0.1.1/24, Area 0
  ! Truncated
FastEthernet1/0 is up, line protocol is up
  Internet address is 10.0.2.1/24, Area 0
  ! Truncated
FastEthernet1/1 is up, line protocol is up
  Internet address is 10.0.3.1/24, Area 0
  ! Truncated
Loopback0 is up, line protocol is up
  Internet address is 192.168.0.1/32, Area 0
  ! Truncated
```

Correct the configuration on R1 so that all interfaces are in Area 1.

```
R1(config)#router ospf 1
R1(config-router)# network 10.0.0.0 0.255.255.255 area 1
R1(config-router)# network 192.168.0.0 0.0.0.255 area 1
```

```
R1#sh ip ospf interface
```

```
FastEthernet0/1 is up, line protocol is up
  Internet address is 10.0.1.1/24, Area 1
  ! Truncated
FastEthernet1/0 is up, line protocol is up
  Internet address is 10.0.2.1/24, Area 1
  ! Truncated
FastEthernet1/1 is up, line protocol is up
  Internet address is 10.0.3.1/24, Area 1
  ! Truncated
Loopback0 is up, line protocol is up
  Internet address is 192.168.0.1/32, Area 1
  ! Truncated
```

The configuration is corrected but R1 still has no OSPF neighbours.

```
R1#sh ip ospf neighbor
R1#
```

Let's check R2 next. 'Show ip protocols' does not show any problems with the configuration, but the absence of neighbor 'Routing Information Sources' shows that it has not formed any adjacencies.

```
R2#sh ip protocols
*** IP Routing is NSF aware ***
```

```
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.0.2
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.255 area 1
    10.1.0.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.0.2      110          00:17:53
  Distance: (default is 110)
```

Check for Layer 1 or Layer 2 issues with 'show ip interface brief'.

```
R2#show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
FastEthernet0/0    10.0.0.2        YES NVRAM  administratively down down
FastEthernet0/1    10.1.0.2        YES NVRAM  administratively down down
FastEthernet1/0    unassigned      YES NVRAM  administratively down down
FastEthernet1/1    unassigned      YES NVRAM  administratively down down
Loopback0          192.168.0.2     YES NVRAM  up up
```

We can see that the OSPF interfaces FastEthernet 0/0 and 0/1 have been administratively shut down. Bring the interfaces up.

```
R2(config)#interface f0/0
R2(config-if)#no shutdown
R2(config-if)#interface f0/1
R2(config-if)#no shutdown
```

The neighbour relationship comes up with R1 (give it a little time to converge) and OSPF routes are received from it. The adjacency with R3 is still down.

```
R2#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.0.1	1	FULL/BDR	00:00:33	10.0.0.1	FastEthernet0/0

```
R2#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C    10.0.0.0/24 is directly connected, FastEthernet0/0
L    10.0.0.2/32 is directly connected, FastEthernet0/0
O    10.0.1.0/24 [110/2000] via 10.0.0.1, 00:07:00, FastEthernet0/0
O    10.0.2.0/24 [110/2000] via 10.0.0.1, 00:07:00, FastEthernet0/0
O    10.0.3.0/24 [110/2500] via 10.0.0.1, 00:07:00, FastEthernet0/0
C    10.1.0.0/24 is directly connected, FastEthernet0/1
L    10.1.0.2/32 is directly connected, FastEthernet0/1
192.168.0.0/32 is subnetted, 2 subnets
O    192.168.0.1 [110/1001] via 10.0.0.1, 00:07:00, FastEthernet0/0
C    192.168.0.2 is directly connected, Loopback0
```

'Show ip protocols' on R3 shows that it is not routing for any networks.

```
R3#show ip protocols
*** IP Routing is NSF aware ***
```

```
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.0.3
  Number of areas in this router is 0. 0 normal 0 stub 0
  nssa
  Maximum path: 4
  Routing for Networks:
  Routing Information Sources:
    Gateway           Distance           Last Update
  Distance: (default is 110)
```

Add network statements for its interfaces.

```
R3(config)#router ospf 1
R3(config-router)#network 10.0.0.0 0.255.255.255 area 0
R3(config-router)#network 192.168.0.0 0.0.0.255 area 0
```

The adjacency with R2 comes up and they exchange routes. There is still no adjacency with R4.

```
R3#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.0.2	1	FULL/DR	00:00:36	10.1.0.2	FastEthernet0/1

```
R3#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O IA   10.0.0.0/24 [110/2000] via 10.1.0.2, 00:00:44, FastEthernet0/1
O IA   10.0.1.0/24 [110/3000] via 10.1.0.2, 00:00:44, FastEthernet0/1
O IA   10.0.2.0/24 [110/3000] via 10.1.0.2, 00:00:44, FastEthernet0/1
O IA   10.0.3.0/24 [110/3500] via 10.1.0.2, 00:00:44, FastEthernet0/1
C      10.1.0.0/24 is directly connected, FastEthernet0/1
L      10.1.0.1/32 is directly connected, FastEthernet0/1
C      10.1.1.0/24 is directly connected, FastEthernet0/0
L      10.1.1.2/32 is directly connected, FastEthernet0/0
```

```

    192.168.0.0/32 is subnetted, 2 subnets
O IA    192.168.0.1 [110/2001] via 10.1.0.2, 00:00:44, FastEthernet0/1
C       192.168.0.3 is directly connected, Loopback0

```

'Show ip protocols' on R4 shows that it has a misconfigured network statement. It is routing for 10.0.0.0/24, but its interfaces have IP addresses in the 10.1.0.0/16 range.

```

R4#show ip protocols
*** IP Routing is NSF aware ***

```

```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.0.4
  It is an autonomous system boundary router
  Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0
  nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.255 area 0
    192.168.0.0 0.0.0.255 area 0
    203.0.113.0 0.0.0.255 area 0
  Passive Interface(s):
    FastEthernet3/0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)

```

Replace the misconfigured network statement. Use a statement which covers the 10.1.0.0/16 addresses (I use 10.0.0.0/8 here which will work).

```

R4(config)#router ospf 1
R4(config-router)#no network 10.0.0.0 0.0.0.255 area 0
R4(config-router)#network 10.0.0.0 0.255.255.255 area 0

```

The adjacency with R3 comes up and they exchange routes. There is still no adjacency with R5.

```

R4#show ip ospf nei

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.0.3	1	FULL/DR	00:00:37	10.1.1.2	FastEthernet0/0

R4#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

```
S*    0.0.0.0/0 [1/0] via 203.0.113.2
      10.0.0.0/8 is variably subnetted, 11 subnets, 2 masks
O IA   10.0.0.0/24 [110/3000] via 10.1.1.2, 00:00:33, FastEthernet0/0
O IA   10.0.1.0/24 [110/4000] via 10.1.1.2, 00:00:33, FastEthernet0/0
O IA   10.0.2.0/24 [110/4000] via 10.1.1.2, 00:00:33, FastEthernet0/0
O IA   10.0.3.0/24 [110/4500] via 10.1.1.2, 00:00:33, FastEthernet0/0
O      10.1.0.0/24 [110/2000] via 10.1.1.2, 00:00:33, FastEthernet0/0
C      10.1.1.0/24 is directly connected, FastEthernet0/0
L      10.1.1.1/32 is directly connected, FastEthernet0/0
C      10.1.2.0/24 is directly connected, FastEthernet0/1
L      10.1.2.1/32 is directly connected, FastEthernet0/1
C      10.1.3.0/24 is directly connected, FastEthernet1/0
L      10.1.3.1/32 is directly connected, FastEthernet1/0
      192.168.0.0/32 is subnetted, 3 subnets
O IA   192.168.0.1 [110/3001] via 10.1.1.2, 00:00:33, FastEthernet0/0
O      192.168.0.3 [110/1001] via 10.1.1.2, 00:00:33, FastEthernet0/0
C      192.168.0.4 is directly connected, Loopback0
      203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C      203.0.113.0/24 is directly connected, FastEthernet1/1
L      203.0.113.1/32 is directly connected, FastEthernet1/1
```

'Show ip protocols' on R5 shows that FastEthernet 0/0 and 0/1 are passive interfaces.

R5#show ip protocols

*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 192.168.0.5

It is an area border router

Number of areas in this router is 2. 2 normal 0 stub 0

nssa

Maximum path: 4

Routing for Networks:

10.0.3.0 0.0.0.255 area 1

10.1.3.0 0.0.0.255 area 0

Passive Interface(s):

FastEthernet0/0

FastEthernet0/1

Routing Information Sources:

Gateway Distance Last Update
Distance: (default is 110)

Reconfigure the interfaces so they will form adjacencies.

```
R5(config)#router ospf 1
R5(config-router)#no passive-interface fastEthernet 0/0
R5(config-router)#no passive-interface fastEthernet 0/1
```

The adjacencies with R4 and R1 come up and they exchange routes.

R5#show ip ospf nei

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.0.4	1	FULL/DR	00:00:31	10.1.3.1	FastEthernet0/0
192.168.0.1	1	FULL/DR	00:00:36	10.0.3.1	FastEthernet0/1

R5#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is 10.1.3.1 to network 0.0.0.0

```
O*E2 0.0.0.0/0 [110/1] via 10.1.3.1, 00:00:50, FastEthernet0/0
      10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O      10.0.0.0/24 [110/2500] via 10.0.3.1, 00:01:05, FastEthernet0/1
O      10.0.1.0/24 [110/2500] via 10.0.3.1, 00:01:05, FastEthernet0/1
O      10.0.2.0/24 [110/2500] via 10.0.3.1, 00:01:05, FastEthernet0/1
C      10.0.3.0/24 is directly connected, FastEthernet0/1
L      10.0.3.2/32 is directly connected, FastEthernet0/1
O      10.1.0.0/24 [110/3500] via 10.1.3.1, 00:00:50, FastEthernet0/0
O      10.1.1.0/24 [110/2500] via 10.1.3.1, 00:00:50, FastEthernet0/0
O      10.1.2.0/24 [110/2500] via 10.1.3.1, 00:00:50, FastEthernet0/0
C      10.1.3.0/24 is directly connected, FastEthernet0/0
L      10.1.3.2/32 is directly connected, FastEthernet0/0
      192.168.0.0/32 is subnetted, 4 subnets
O      192.168.0.1 [110/1501] via 10.0.3.1, 00:01:05, FastEthernet0/1
O      192.168.0.3 [110/2501] via 10.1.3.1, 00:00:50, FastEthernet0/0
O      192.168.0.4 [110/1501] via 10.1.3.1, 00:00:50, FastEthernet0/0
C      192.168.0.5 is directly connected, Loopback0
O      203.0.113.0/24 [110/2500] via 10.1.3.1, 00:00:50, FastEthernet0/0
```

All routes should now be available on all routers.

R1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

```
10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C    10.0.0.0/24 is directly connected, FastEthernet0/0
L    10.0.0.1/32 is directly connected, FastEthernet0/0
C    10.0.1.0/24 is directly connected, FastEthernet0/1
L    10.0.1.1/32 is directly connected, FastEthernet0/1
C    10.0.2.0/24 is directly connected, FastEthernet1/0
L    10.0.2.1/32 is directly connected, FastEthernet1/0
C    10.0.3.0/24 is directly connected, FastEthernet1/1
L    10.0.3.1/32 is directly connected, FastEthernet1/1
O IA 10.1.0.0/24 [110/2] via 10.0.0.2, 00:03:56, FastEthernet0/0
O IA 10.1.1.0/24 [110/3] via 10.0.0.2, 00:03:56, FastEthernet0/0
O IA 10.1.2.0/24 [110/4] via 10.0.0.2, 00:03:56, FastEthernet0/0
O IA 10.1.3.0/24 [110/1503] via 10.0.0.2, 00:03:56, FastEthernet0/0
192.168.0.0/32 is subnetted, 3 subnets
C    192.168.0.1/32 is directly connected, Loopback0
O IA 192.168.0.3/32 [110/3] via 10.0.0.2, 00:03:56, FastEthernet0/0
O IA 192.168.0.4/32 [110/4] via 10.0.0.2, 00:03:56, FastEthernet0/0
O IA 203.0.113.0/24 [110/4] via 10.0.0.2, 00:03:56, FastEthernet0/0
O*E2 0.0.0.0/0 [110/1] via 10.0.0.2, 00:03:56, FastEthernet0/0
```